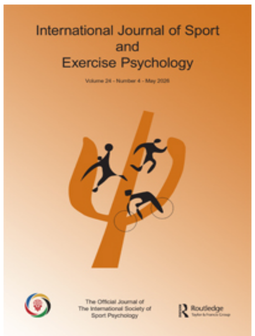

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REVIEW ARTICLE

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Using video-notational analysis to examine soccer players' behaviours

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ABSTRACT

The purpose of numerous scientific disciplines is to comprehend human behaviour. However, much of human behaviour that is understood to date occurs in artificial laboratory settings. Although this approach allows researchers to control variables, it is limited in understanding human behaviour in natural, fast and high pressure performance environments like soccer. Video-notational analysis could address these limitations by providing behavioural observations, following constructionist perspectives that argue that understanding behaviours in context – relative to environments and relationships – is fundamental. Accordingly, this conceptual paper explores the use of video-notational analysis to examine players' behaviours in soccer, particularly emotions, visual exploratory activity (VEA) and intra-team communication. These topics were selected by the technical staff of Ajax AFC as they could provide immediate video inputs to the team and individual players about their psychological performance. The paper reviews previous literature on the use of video-notational analysis in understanding these players' behaviours and its potential contributions to the field of sport psychology and performance analysis for future research, integrating emerging technologies such as artificial intelligence (AI), machine learning as well as practical implications and reflections.

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Behavioural analysis; notational analysis; performance analysis; sport psychology; video analysis

During my time at the soccer club Ajax AFC, I (S.C.) held various roles, including youth coach, performance video analyst and researcher, primarily focusing on sport psychology research. When Erik ten Hag became the head coach of the first team, he expressed the desire to integrate psychological topics into the performance video (tactical and technical) analysis. As a result, I joined the first team's technical staff with the role of developing performance video analysis that incorporated sport psychology elements. This led to the creation of the video behavioural analyst role. This role was enriched with the feedback

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from the technical staff and the integration of the latest research findings in the field of sport psychology. In particular, the position of the video behavioural analyst primarily involved the collaboration with the sport psychologist, performance video analyst, head coach, and assistant coaches, by providing video footages of specific team and individual behaviours (e.g., behavioural patterns) as well as data from specific topics (e.g., the number of intra-team communications among players). This (video)information was used to enrich team conversations and individual player feedback, particularly during team presentations (e.g., post-match analysis presentations) or individual evaluations, which often were held by the head coach and the sport psychologist. This method helped the players to become more aware of certain behaviours which they were often unaware of.

This (video) data information was also of interest to answer scientific questions besides the applied questions described above used to conduct scientific research, which later became part of my PhD study. Hence, the present manuscript details how the tool of video-notational analysis can meet the standards of scientific rigour to answer interesting questions in sport psychology. Here we argue that this approach has the substantial advantage to measure behaviour in the very context that you are trying to understand/modify/predict, which has been an important problem in the behavioural sciences (e.g., Neisser, 1976).

In recent years, the video-notational analysis research has seen a substantial growth in the literature. However, the majority of studies are designed such that researchers either use match videos (e.g., recorded games), often found on the internet, to conduct their studies; or they receive specific requests from clubs to carry out research by analysing provided footage. Clubs typically allow researchers to visit them for limited periods and times. Therefore, it is rare for a researcher to work full-time within a soccer club as an integral member of the technical staff, fully involved in the team's processes. Further, coaches are often employed by the clubs on a short-term basis and are primarily focused on achieving immediate results. Consequently, it is even less common for a video (behavioural) analyst to combine the work directly utilised by the technical staff with the creating of material suitable for research. We believe that this represents an unique situation, providing a valuable opportunity for the sport psychology community to not only gain deeper insights and practical applications in professional sports, but exploit and enhance tools used in this area – in this case video-notional analysis – for scientific research. Therefore, we decided to write this *conceptual* paper (see Jaakkola, 2020) to advocate the development of video behavioural analysis from both a practical and scientific perspective.

This conceptual paper follows the structure of theory synthesis. It first outlines the importance of examining players' behaviours through video-notational analysis for both practice and research. It then focuses on the phases of developing behavioural video-notational analysis (e.g., the software criterion). Thirdly, it examines the importance of these analysis also for research, highlighting the benefits for both researchers and practitioners. Next, the second section (i.e., *behavioural categories*), identifies the possible behaviours to analyse and include in the video notational analysis. These behaviours were identified by the technical staff, with inputs also from the first author, which were based on the current video-notational analysis literature. Thus, the topics examined in this paper included intra-team communication, emotions, and visual explorative activity

(VEA). It is important to note that these topics were selected by the technical staff from the perspective that this video information could potentially provide immediate input to the team or individual players, with the aim of improving performance. By using a narrative approach (Furley & Goldschmied, 2021), we aimed to first detail our approach and integrate current literature and exemplify how this can be used to advance understanding of difficult to measure behaviour in situ before we showcase fruitful avenues for future research.

Using video-notational analysis to examine players behaviors

The aim of numerous scientific disciplines is to understand human behaviour. However, much of human behaviour that is understood to date occurs in artificial laboratory settings (e.g., artificial experimental setups) (Neisser, 1976; Van der Kamp et al., 2008; Williams et al., 1999; Williams & Abernethy, 2012). Although this approach allows researchers to control variables – which is an important factor in scientific research – it is limited in understanding human behaviour in natural, fast and high pressure performance environments like soccer. Thus, often sports research has been criticised of using convenient measures instead of measures representative the real-life settings (Meredith et al., 2018). In contrast, qualitative researchers, particularly from constructionist perspectives, argue that understanding behaviour in context – relative to environments and relationships – is fundamental. These perspectives emphasise that human behaviour is best understood within the complexity of real-world interactions and settings (Aguinaldo, 2004). Similarly, researchers have also shown that sport psychology research purporting to understand athletes' behaviours rarely directly examines those actual behaviours, rather indirect measures are relied upon to say something about behaviour (e.g., questionnaires and interviews) (Andersen et al., 2007). Compounding this challenge, sport researchers face the time-intensive nature of planning studies, while coaches demand immediate solutions to multifaceted problems often lacking clear scientific definitions (Sands, 1998).

Video and notational analysis emerge as a promising solution, bridging the gap between controlled laboratory studies, qualitative research, and the dynamic demands of the real-world of sports. The use of video-notational analysis aligns with ecological psychology and representative design theories, emphasising the study of naturalistic behaviours. Specifically, in the context of sports performance, "representative design" entails the simulation of conditions akin to those encountered by the players in the real competition (Brunswik, 1956; Gibson, 1979). As a result, representative design analysis gained prominence in sports psychology research for its ecological validity (Araújo et al., 2007). However, while video-notational analysis witnessed substantial growth in the literature in recent years, their application in applied settings remains still an unstudied area, necessitating further development and research to fully harness its potential (Carling et al., 2005, 2009; Hughes, 1997; Hughes & Franks, 2004; James, 2006; Wright et al., 2014). The next section outlines the practical tools involved in using video-notational analysis (e.g., audio-video analysis). It then discusses considerations for selecting appropriate software, such as functionality and ease of use. This information is valuable for practitioners and researchers, providing guidance on aspects like coding capabilities during analysis to enhance the effectiveness of their work or research.

The practical use of video-notational analysis

Video-notational analysis involves the use of several tools. It includes the recording of an event and analysing the footage either live or after the event has occurred. If note-taking is conducted during these analyses, it is referred to as written analysis (i.e., notational analysis). The events captured in coded videos can also be examined through artificial intelligence (AI) tools, which can facilitate automatic feedback from automated analyses (Hughes & Franks, 2004). Moreover, audio recordings may be incorporated during the analysis. For example, when analysing verbal communication, audio recordings may complete the video footage. However, examining communication is typically challenging, particularly for verbal communication. The presence of the audience and the on-field noise from various sources (e.g., technical staff, referees and ongoing exchanges between team members) makes it intricate. Additionally, sporting federations typically prohibit the use of microphones for players to wear. Consequently, the majority of studies and practitioners predominantly rely on video analysis and are largely confined to nonverbal communication analysis. Thus, behavioural video-notational analysis did not consider studies conducted with audio recordings, such as observations of coaching styles (Becker & Wrisberg, 2008) and verbal communication (Lausic et al., 2009, 2014, 2015; LeCouteur & Feo, 2011). Neither in practice, despite there are emerging companies that could facilitate this process, for example Vokalo[1] and Coach Whisperer.[2]

There has been a notable increase of video analysis software tools in recent years (e.g., Spiideo, Angles, and Dartfish). Irrespective of the software preference of an user, the software should adhere to seven criteria. (1) *User-friendliness*. The video, observational instrument and recorded data are all presented concurrently on the screen, accessible without needing specialised computer expertise. The software used to tag moments, in addition to being available on a computer, would benefit for both practitioners and researchers if they were also accessible as a user-friendly app for tablets and smartphones, especially for tagging live moments. Teams often have individuals in the technical staff who work as both coaches on the field and video analysts. As a result, they frequently sit with the technical staff on the bench during matches and can tag key moments that they would like to review or show to the players within a limited timeframe, such as during halftime. Additionally, in some research focused on analysing live moments (e.g., Fransen et al., 2014), a tablet or smartphone may be a more suitable tool than a computer, making these devices valuable options for researchers. (2) *Possibility of customising (ad hoc) the observational tool*. Once set up, the instrument is ready to use, allowing users to define and customise criteria and categories to suit their needs. This is especially common in elite teams, where the technical staff consists of coaches and analysts with specific areas of focus. For example, a video behavioural analyst may require a customised observation tool, as standard tools for analysing tactical and technical moments typically do not include behavioural factors. Another example is with goalkeeper coaches, who may want to create their own specific tags to analyse goalkeeping performance more effectively. For research, too. Researchers could conduct notational analysis on an observation sheet and then report the data on a computer. However, having a customised observation tool on a computer or a tablet would make data processing faster. (3) *Automatic recording of the time at which an action takes place*. Upon registration, the event's time and duration are automatically recorded. (4) *Direct linkage between video and events*. Recorded data is linked to specific video frames, allowing users to easily access and review observed behaviours within the session. (5) *Less time required for observation*. Video and data registering are controlled through simple mouse clicks or touches on the screen. (6) *Higher data quality*. The presence of well-defined categories and the elimination of the need for data transcription contribute to error prevention. (7) *Simultaneous data analysis*. The analysis tool performs frequency and duration analyses and allows users to regroup data by relating variables or features to each other (Brewer & Jones, 2002; Castellano et al., 2008). As video-notational analysis tools are implemented and used in practice, researchers can leverage these tools to conduct research. The following section discusses observational methods, providing foundational insights for effectively analysing and interpreting behaviours.

The use of video-notational analysis for research

Heyman et al. (2014) categorised behavioural observation into four types: quasi-naturalistic observation, analogue observation, experimental manipulation and naturalistic observation. Quasi-naturalistic observation combines naturalistic and controlled settings, often used in studies like small-sided games (see Caso & van der Kamp, 2020). Analogue observation creates controlled environments to study behaviours and interactions (Heyman, 2001). Experimental manipulation, vital in controlled lab settings, allows for the alteration and observation of specific factors influencing behaviour (Prentice & Miller, 1992). Finally, naturalistic observation involves qualitative or quantitative approaches and it provides the benefit of possessing high ecological validity (Araújo et al., 2007). It provides objective data on the occurrence rates of specific behaviours exhibited by an individual (Carling et al., 2009). Additionally, these data can be used to investigate also the mechanisms underlying social interactions, for example, examining the interactions between players and their leadership styles (Carling et al., 2009; Fransen et al., 2014; Gray et al., 2008; Heyman et al., 2014).

Behavioural coding plays an important role in research and it comes in two main types: topographical coding systems, which measure the occurrence of behaviours; and dimensional coding systems, which measure the intensity of behaviours. The selection of a coding system depends on the study's specific objectives. Ideally, researchers can use an already existing system that aligns with their needs from previous literature (Kerig & Baucom, 2004; Kerig & Lindahl, 2001). However, this is not always the case, and researchers may choose a coding system that involves different theoretical considerations (Bakeman & Gottman, 1997). Therefore, designing coding systems may begin with several hypotheses, followed through various phases. One phase involves periods of observation, note-taking and literature review to understand the practical aspects. Furthermore, training observers is fundamental to ensure consistency. The training involves familiarising coders with the coding concepts, providing detailed manuals and progressing from basic to challenging examples. Coders should reach an agreement on coded behaviours to prevent error variance until the interrater agreement reaches the sufficient level (Cohen, 1969). While the previous sections outlined the steps for setting up video behavioural analysis, including considerations such as software features, the next section focuses on the specific topics addressed in the analysis.

Behavioural categories for coding

Which psychological topics could be included?

There are numerous performance behaviours that could be examined. As a result, it is important to ascertain which ones to examine and the rationale for doing so. The key to effectively utilising video-notational analysis is to determine which information is important and whether it is meaningfully associated with the team and individual players' performance. For example, the coach and analyst may wish to identify performance errors and weaknesses to recommend areas for improvement (Carling et al., 2005). This can be achieved by both collecting video footage and transforming it into analyzable data. Thus, as mentioned in the introduction, together with the technical staff and based on the current literature, the following topics were examined: intra-team communication, emotions, and VEA. From practice, video-notational analysis can be utilised to guide team preparation and performance at different stages, see Figure 1. Moreover, the behavioural information gathered not only enriches knowledge of performance, both on an individual and team level, but it could also provide useful insights for scouting and opponent analysis purposes. These ideas are shared in the sections dedicated to practical implications and future research. Before that, each topic will be introduced to the readers through a narrative review. More specifically, the narrative review will follow the structure of first introducing the topic and thereafter describing how it has been studied via video-

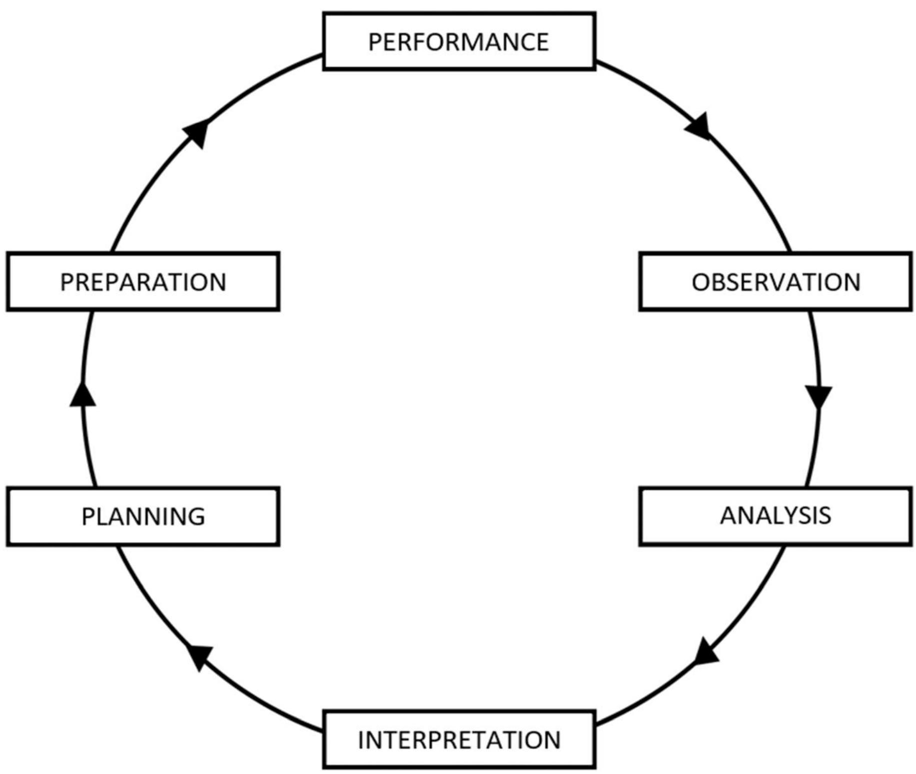

Figure 1. The different video analysis phases.

notational analysis. The reason behind this is that this mimics the process of how behavioural video analysis was developed at Ajax Amsterdam.

Intra-team communication

The importance of effective intra-team communication has been widely recognised by coaches, athletes, analysts and sport psychologists. Some experts even argued that it is better for a team to communicate too much, rather than too little (Lubbers, 2005). Encouraging players to engage in more communication aim to enhance coordination and cooperation among the team members. While cooperation involves individuals working together and supporting each other for the team's benefit, coordination refers to the ability of team members to synchronise their actions to achieve a common goal (Tamboer et al., 2016). Both cooperation and coordination are vital aspects of team performance and contribute to team dynamics, where communication, trust and shared goals among teammates play crucial roles in improving team performance (McGarry et al., 2002).

To assess intra-team communication within teams, several researchers used video-notational observations of matches (Durdubas et al., 2021; Halldorsson, 2018; Kraus et al., 2010; Moesch et al., 2015). Kraus et al. (2010) examined the correlation between tactile communication (e.g., high fives and shoulder taps) and cooperation with match outcomes during elite basketball matches. The findings indicated that the frequency of touch among players was a predictor of performance. A similar study investigated intra-team communication in relation to team success in elite volleyball (Durdubas et al., 2021). Researchers examined various types of communication such as instructional and tactical gestures. The findings indicated that successful teams exhibited a greater overall number of behaviours and used significantly more instructional and supportive gestures compared to less successful teams. Furthermore, successful teams displayed more frequent use of instructional gestures in the games won and more supportive behaviours in the games lost. Additionally, nonverbal intra-team communication was examined during elite team handball matches (Moesch et al., 2015). The research focused on gestures like clapping hands, thumbs up, high fives and others. The results demonstrated that winning teams tended to exhibit a more consistent behavioural pattern in nonverbal communication during the matches, regardless of the time-point or score. Therefore, research suggests that intra-team communication, positively impacts team performance. Kraus et al. (2010) linked tactical communication to better performance. Durdubas et al. (2021) found that successful teams displayed more instructional and supportive gestures. Moesch et al. (2015) highlighted that winning teams showed more stable nonverbal communication. These findings imply that effective intra-team nonverbal communication and positive nonverbal gestures are key for enhancing team performance and therefore part of the behavioural video notational analysis.

Emotions

Which emotions should be included?

Emotional expressions

Emotions influence many aspects of human behaviour (Furley et al., 2012, 2023). They were shown to impact human performance in a variety of domains (Eccles et al., 2011).

Throughout the previous century, the two most common acknowledged theories were the basic emotion and the dimensional theories. However, these theories were deemed contradictory (Barrett & Russell, 2015; Lindquist et al., 2013). The distinction between the two lies in whether emotions are recognised as distinct entities or as an autonomous dimension (Bestelmeyer et al., 2017). According to the basic emotion view, humans are proposed to have universal basic emotions with distinct bodily expressions (Ekman & Rosenberg, 1997): for example, fear, anger, joy, sad, contempt, disgust and surprise (Ekman, 1992). Although the universal expression view is controversial, most emotion theorists agree that emotions are often expressed in facial and bodily expressions and therefore it is possible to analyse them using video footage.

The focus of the video-notational analysis within the performance context is primarily on instantaneous emotions, particularly following events such as goals (e.g., players blaming each other in the aftermath of opponent goals). Bodily expressions of emotions are sometimes divided into micro and macro expressions and these expressions, with the former being indicative of "true" emotions. Micro expressions are being of no more than 500 ms (Ekman & Friesen, 1978; Yan et al., 2013), while macro expressions have usually a duration between 500 ms and four seconds. With the use of cameras and software capable of slow-motion analysis, emotional expressions among players could be examined to identify psychological factors such as players' reactions after certain events or conflicts between players and their reactions.

These emotional expressions serve as indicators, conveying valuable information (Shariff & Tracy, 2011). This information forms the basis for individual emotional responses, the deduction of expected performance standards or the stimulation of specific behaviours. Notably, the perception of emotions exhibited by baseball pitchers, whether anger or happiness, was studied to influence the expectations of pitch speed, difficulty and accuracy (Cheshin et al., 2016). However, these expectations did not consistently align with actual performance metrics, suggesting that the assessments of players' potential based on emotional expressions may not accurately reflect their true performance. In soccer penalty shootout scenarios, research has shown that players expressing pride were rated higher in confidence, composure, relaxation, happiness and reduced shame; while those displaying shame were rated lower, indicating increased stress and shame (Furley et al., 2015). Teammates observing pride expressions anticipate increased confidence, control, pride, happiness and enhanced performance. Conversely, when viewing these expressions as an opposing player, the heightened feelings of shame are linked to reduced performance. This suggests that pride expressions positively impact teammates' feelings. Hence, players' emotional expressions are not always isolated; they are influenced by match outcomes, social interactions, audience and possibly by other psychological components (Crivelli et al., 2015).

Therefore, facial expressions, particularly micro expressions, provide important cues to players' emotions and reactions during critical moments. Thus, emotional expressions may play a crucial role in shaping players' thoughts, emotions, behaviours and performance and can therefore be and interesting variable for behavioural video analysis.

Individual and team emotions

Emotions in teams are a complex interplay of individual and collective psychological mechanisms. For instance, group-based emotions may be particularly influential in shaping team emotions. Several studies explored the interplay of different emotional dynamic mechanisms in this context, including group-based emotions, collective emotions and emotional contagion (Cotterill et al., 2020; Moll et al., 2010; Rumbold et al., 2012; Totterdell, 2000; van Kleef et al., 2019; Wergin et al., 2019; Wolf et al., 2018).

Group-based emotions are individual emotional responses triggered by events relevant to a specific group (e.g., players arguing). The degree to which one identifies with the group and interprets the event are critical factors shaping these emotions. For example, a player might experience group-based anger following their team's loss in a game (Goldenberg et al., 2014, 2020; Mackie et al., 2000). Conversely, collective emotions are emotions experienced by a group of individuals in response to the same situation or event. These emotions are different from group-based emotions, which refer to an individual's emotional experience in response to a group-related event. They depend on group members responding to a situation in the same way and are influenced by social identity theory and appraisal theories of emotion. When a specific collective emotion is strong, individuals who identify as part of that group are more likely to personally experience the same emotion in a group-based form. In the context of soccer, collective emotions and perceptions about teammates following competitive matches are strongly linked to individuals' group-based emotions (Goldenberg et al., 2020; Lazarus, 1991; Tajfel & Turner, 1986).

Finally, emotional contagion is the process by which a person or group influences the emotions of another person or group through the induction of emotion states and behavioural attitudes. This process is influenced by mimicry and afferent feedback, and is linked to group-related emotions (Barsade, 2002; Gump & Kulik, 1997; Hatfield et al., 1993; van der Schalk et al., 2011). For instance, Moll et al. (2010) studied through video-notational analysis the post-shot nonverbal behaviours in penalty kicks from the player's post-kick response within five seconds of taking the penalty. Post-shot responses were judged using a coding scheme developed from a nonverbal behaviour coding system (Tracy & Robins, 2008), associated with pride such as head tilting, smiling, arms raised, fists made, hands on hips and torso expansion. The findings showed that celebratory actions were exhibited by 66% of the players, with the majority showing behaviours associated with pride. Interestingly, those who demonstrated celebratory behaviours were more likely to be part of the winning team in the penalty shootout, supporting the hypothesis that expressions of pride after a successful penalty kick can enhance team performance in shootouts.

Therefore, the importance of emotions in both individual and team contexts has been studied by researchers. Research has highlighted the complex interplay between individual and collective emotional dynamics within teams, often examining group-based emotions, collective emotions, and emotional contagion separately. Group-based emotions, influenced by social identity and event appraisal, are experienced in response to group-related events, such as a player feeling group-based anger after a loss. Collective emotions, on the other hand, are experienced by a group of individuals in response to the same situation and are linked to perceptions about teammates and group identity. Emotional contagion, a process by which emotions are transmitted within a group, is influenced by mimicry and afferent feedback, contributing to group-related emotions. Thus, these behaviours could be included in video-notational analysis, allowing analysts to examine group-based emotions, collective emotions, and emotional contagion, for example, by tracking specific emotions for the team and/or individual players.

Visual exploratory activity

Soccer is a sport that demands synchronised actions, such as coordinated team pressure, creating opportunities for passing and implementing tactical plans. For instance, diamond-shaped movements involve players strategically positioning themselves on the field to form triangles, thereby enabling passing opportunities for their teammates (da Costa et al., 2010). To gather the necessary tactical information for this, players can review videos, for example during team meetings. However, during the actual match, constant communication between players is essential to effectively coordinate their actions based on their understanding of the game environment. In certain situations, players may even coordinate their actions without explicit communication by utilising eye contact or recognising opportunities for action (i.e., affordances) (Gibson, 1966, 1979).

This coordination process can be achieved through exploration, which involves a continuous and active process of individuals gathering and absorbing information while in action, known as VEA or also "scanning". VEA was originally defined by Jordet (2005) as "a body and/or head movement in which the player's face is actively and temporarily turned away from the ball, seemingly with the intent of observing teammates, opponents, and other environmental elements or events relevant to the upcoming action with the ball" (p. 143). Studies have shown that VEA engaged in prior to receiving the ball is positive related to successful performance (Aksum et al., 2021a, 2021b; Caso et al., 2023, 2024a; Eldridge et al., 2013; Jordet et al., 2013, 2020; McGuckian et al., 2017, 2018, 2020; Phatak & Gruber, 2019; Pokolm et al., 2022) and other aspects of the game such as affecting the opponent's team's pressure (Aksum et al., 2021a, 2021b; Eldridge et al., 2013; Jordet et al., 2020; Pokolm et al., 2022). Moreover, recent research by Caso et al. (2023) demonstrated that players who actively explore their surroundings during the penultimate pass before receiving the ball tend to achieve greater success in their performance. This suggests that VEA also serves as an anticipation factor, allowing players to recognise early possibilities for action in a given situation.

Therefore, it seems that VEA plays a dual role in enhancing soccer players' performance. It not only aids in gathering crucial information for the upcoming play but also facilitates anticipation, enabling players to identify potential actions ahead of time. This skill set allows players to make more informed decisions and execute their strategies with precision on the field and these behaviours are observable via video-notational analysis.

Research gaps

Video-notational analysis, highlighted by its emphasis on naturalistic behaviour in sports performance contexts (Brunswik, 1956; Gibson, 1979), serves as a link between controlled laboratory studies and the dynamic realities of sports. It represents a valuable tool for future research in soccer. For example, within intra-team communication dynamics, future studies could examine the relationship between intra-team communication (verbal, nonverbal, and both) patterns and team success (Durdubas et al., 2021; Eccles & Tenenbaum, 2004; Halldorsson, 2018; Kraus et al., 2010; Lausic, 2005; McLean et al., 2021; Moesch et al., 2015). Specific variables like tactical communication could be examined as indicator for effective team coordination and success (Durdubas et al., 2021). Further, utilising AI-driven algorithms to analyse (possibly live) large datasets of communication patterns could provide more detailed analysis. For instance, analysing intra-team communication during a match may reveal patterns corresponding to goal-scoring opportunities and periods of low intra-team communication associated with goals conceded. Hence, intra-team communication algorithms may be developed to predict goal opportunities, similarly to the expected goal (xG) model (Umami et al., 2021). Moreover, intra-team communication could be studied in terms of whether it is influenced by psychological factors, such as home and away games (Pollard, 2006; Zheng et al., 2025) or the "underdog" effect (Frazier & Snyder, 1991). This could be done by automatic AI analysis that may help to identify patterns and correlations within the intra-team communication data. For example, exploring specific individual players' intra-team communication patterns and potential connections with psychological tests could provide a deeper understanding of the role of communication (Aguiar et al., 2010).

The integration of AI into emotional analysis might also be a promising avenue for future application. AI algorithms, when applied to video footage, have the potential to enhance the accuracy and efficiency of emotion recognition such as distinctions between micro and macro expressions. This could increase the understanding of how emotions unfold in real-time during live matches. Consequently, future research could explore methodologies that incorporate AI-driven emotional analysis to capture the intricacies of players' emotional moments.

In the domain of VEA, future studies may examine aspects related to talent identification. In a study involving U17 and U19 elite players, Pokolm et al. (2022) found that players with more national team matches demonstrated higher VEA. Aksum et al. (2021b) also observed in the finals of the 2018 UEFA European Championship that U19 players exhibited more scans than U17 players. Furthermore, McGuckian et al. (2020) found that U23 elite players had higher head turn frequencies in the exploration phase compared to U13 players in the Footbonaut,[3] indicating higher VEA among older youth players. Thus, VEA could be a reliable indicator for talent identification, with potential research comparing VEA patterns of players transitioning into professional careers. Moreover, age-related dimensions of VEA could be explored through longitudinal investigations, identifying important periods for the development of perceptual-motor skills. However, it is important to first examine the potential influence of intra-team communication on VEA to determine at what (certain) degree VEA is affected by intra-team communication (e.g., when teammates shout to each other "man on"). By utilising AI algorithms, researchers could automatically analyse VEA especially among youth players. Additionally, AI-driven longitudinal investigations could track the development of VEA over time, identifying important periods for the acquisition of perceptual-motor skills.

To date, VEA research has primarily focused on ball possession (Aksum et al., 2021a, 2021b; Caso et al., 2023; 2024a; Eldridge et al., 2013; Jordet et al., 2013, 2020; McGuckian et al., 2017, 2018, 2020; Phatak & Gruber, 2019; Pokolm et al., 2022). Consequently, the defensive phase of VEA remains largely unexplored. Investigating VEA in the defensive phase could examine differences across defensive performance levels. Finally, several companies (e.g., Be Your Best[4]) are currently utilising virtual reality (VR) to potentially enhance VEA. Thus, studying the potential effects of VR training on VEA during training session or matches through video-notational analysis could be another studying area (Rojas et al., 2020; Wirth et al., 2021).

However, for certain topics and gestures, future studies should first examine the reliability of video-notational analysis. It is true that in video-notational analysis studies, coders reach an agreement on coded behaviours to minimise error variance, ensuring that interrater agreement reaches a sufficient level. Nevertheless, there are topics or gestures where coders may have differing interpretations. For example, the assumptions of video-notational analysis in examining in-match communication (both verbal and nonverbal) were studied by Caso et al. (2024b). This case study, conducted with an elite U19 soccer team, analysed the quantity and types of communication under two conditions: video alone and video combined with audio. Results revealed that the quantity of communication recorded by the two methods did not correlate. Weak differences were found in the distribution of communication types between the two methods, although no single difference was particularly prominent. Additionally, moderate to strong correlations between the two methods were observed for giving indications and negative communication, while no correlations were found for other types. Therefore, some gestures may be easier to analyse using video-notational analysis, such as high-fives between players. However, other gestures or behaviours, like nonverbal intra-team communication or VEA, require more careful study. For instance, this could involve using multiple cameras positioned at various angles around the soccer pitch to ensure a detailed coverage (e.g., each player could be tracked by a single camera).

Practical applications and reflections

The practical implications of behavioural video-notational analysis include a wide range of applications designed to optimise player performance and inform coaching strategies and methodologies. By leveraging data from the analysis of players' behaviours and psychological components, practitioners can make more informed decisions that contribute to a team's success. Additionally, integrating AI application provides new opportunities, allowing practitioners to automate the processing and interpretation of vast amounts of video data, thereby enhancing the efficiency and accuracy of behavioural analysis.

Analysing communication dynamics is important for understanding whether players interact among each other and how, involving both verbal and nonverbal exchanges during matches (McLaren & Spink, 2018; Sullivan & Feltz, 2003). For instance, examining intra-team communication serves as a tool for team goal-setting and players' development, providing inputs into the individual behaviours and team dynamics. Players could have individual communication frequency targets which they have to achieve per game. By examining gestures like high fives and tactical instructions (Durdubas et al., 2021; Kraus et al., 2010), analysts can collect important information into the players' needs and tailor strategies. For example, during a UEFA Champions League match, Erik ten Hag and the technical staff (of Ajax AFC) assigned the specific goal to a few players considered team leaders: if the team conceded a goal, their task was to motivate their teammates with positive gestures. In that match, analysing the data on positive gestures such as clapping, the team recorded one of its highest scores compared to other matches during the season. Moreover, studying communication behaviours influenced by psychological factors, such as home or away games (Pollard, 2006), match results and "underdog" effects (Frazier & Snyder, 1991) and "closed-door" matches (Zheng et al., 2025), enables the practitioners to monitor and optimise the team performance in different conditions. This information shared to the players (e.g., via data reports) increases players' awareness on how psychological factors influence their behaviours and therefore performance. For example, there was a noticeable difference between matches played with closed doors during the COVID-19 season and those played with the audience in the intra-communication data. Specifically, positive gestures such as clapping and thumbs-ups were higher when an audience was present. Additionally, leveraging machine learning and AI technologies to automate communication analysis allows practitioners to efficiently identify patterns in verbal and nonverbal interactions among players during matches. Therefore, machine learning algorithms could also discern nuanced communication actions providing practitioners with more detailed data into team dynamics and facilitating interventions to improve communication effectiveness and foster better team coordination.

The integration of video-notational analysis in studying VEA provides practical implications by examining players' behaviours in their natural settings (Araújo et al., 2007), supporting applications such as video scouting, opponent analysis and technologies like VR training (Eldridge et al., 2023; Jordet et al., 2013; Rojas et al., 2020; Wirth et al., 2021). Further, it may help talent identification and recruitment by examining players' behaviours and their decision-making, such as analysing players' VEA before ball possession. In opponent analysis, it could be used to identify weaknesses and counter strengths of the opponent teams, for example, defenders who are less skilled in performing VEA. Finally, several soccer clubs use VR for training VEA skills (Wirth et al., 2021). Video-notational analysis could be used to examine whether the players trained in VR, improved their VEA and consequently their performance in real-life settings (e.g., during matches).

Machine learning algorithms are capable of tracking and analysing complex patterns in visual scanning strategies. By automating the identification and classification of VEA, coaches could pinpoint players with particularly high or low VEA skills and tailor training programmes to develop and/or optimise their existing abilities. Automatic analysis may independently assess video databases and generate players' VEA ranking for talent scouts, utilising public video footage or recordings from routine youth matches within the soccer clubs. Further, machine learning-driven analysis enables coaches to track players' VEA patterns over time, facilitating personalised development plans and optimising player performance based on individual strengths and weaknesses.

Several coaches have emphasised the significance of emotions. One such coach was the "legendary" basketball coach John Wooden. He did not believe in emotional "outburst" on the basketball court. He firmly believed that emotion acted as an adversary and he underscored the importance of players maintaining a balanced and composed manner, discouraging to demonstrate extreme emotions, be they positive or negative, at the end of a game (Ermeling, 2012). Excessive celebrations and expressions of discouragement were actively discouraged by him, who preferred a measured and controlled approach (Wooden & Jamison, 2003, 2005; Wooden & Tobin, 1972). Therefore, understanding the emotional dynamics among players offers important opportunities for managing emotions and optimising performance. For example, by utilising video-notational analysis to capture emotional expressions, coaches can identify moments of heightened stress, frustration or confidence during matches (Furley et al., 2015). This information enables tailored interventions to regulate emotions and promote a positive team environment conducive to peak performance. This can be applicable also in exploring mental toughness that helps coaches to identify behaviours and characteristics associated with resilience and performance under pressure (Diment, 2014). By recognising instances of mental toughness displayed by players during matches, coaches could cultivate a mentally resilient team culture and implement targeted training programmes to enhance players' psychological readiness for competition. Finally, the incorporation of AI may contribute to the development of personalised interventions aimed at enhancing mental toughness. Coaches and technical staff could gain rapid access to specific video footage (e.g., specific emotional moments between two players).

Conclusions

As the field of sport psychology continues to evolve, video-notational analysis stands out as a key area for future exploration, offering opportunities to expand the understanding of human behaviour in soccer and its implications for performance optimisation and skill development. By embracing the multifaceted nature of player behaviours and the psychological dimensions of performance, video-notational analysis is a valid tool for advancing the field of sport psychology and performance analysis, ultimately benefiting coaches, players, analysts, and researchers in the soccer industry. Analysts can gain information beyond standard tactical and technical assessments, allowing them to analyse individual and team patterns from a psychological-behavioural perspective. This allows also to have interdisciplinary collaboration with experts in sports psychology, physiology, and related fields enriches the analysis process, contributing to holistic training programme design. However, it is important for future studies to analyse the reliability of video-notational analysis for certain behaviours or gestures, as outlined in the study by Caso et al. (2024b).

During the developmental phases of behavioural video-notational analysis, practitioners found intra-team communication, emotions, and VEA to be the most apparent topics that could be analysed via video for quick practical implications. However, there are other potential topics that could be included in video analysis, such as mental toughness (Diment, 2014), creativity and variability in exercises like small-sided games (Caso & van der Kamp, 2020; de Joode et al., 2023), particularly concerning training periodisation (Clemente et al., 2014; Los Arcos et al., 2017) that practitioners and researchers may take in consideration. Finally, emotions are also shaped by coaches who can positively or negatively influence team performance. For example, van Kleef et al. (2019) found that players base performance inferences on coaches' emotional expressions, with coach anger linked to poor performance and increased player anger, while coach happiness correlated with good performance and player happiness. Allan and Côté (2016) showed that composed emotional expressions from coaches foster prosocial behaviours and reduce antisocial behaviours in athletes. Additionally, Staw et al. (2019) found that moderately intense negative emotions expressed during half-time speeches led to greater effort and improved team performance. Therefore, video-notational analysis could also be used to help coaches become more aware of their emotions and behaviours.

Notes

1. https://vokalo.io/
2. https://coachwhisperer.de/
3. The Footbonaut is a football training machine which fires balls at different speeds and trajectories at players, who must control and pass the ball into a highlighted square.
4. https://www.beyourbest.com/

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No potential conflict of interest was reported by the author(s).

Data availability statement

There is no data to be shared as this is a narrative review.

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